Lab Window Function

Name: Lê Hồng Anh
MSV: 11247126

Lab 1: Film Ranking

Result

Top 3 longest films per category (using `DENSE_RANK()` with `PARTITION BY` and CTE):

Query | Query History

```sql
WITH RankedFilms AS (
    SELECT
        c.name AS category_name,
        f.title,
        f.length,
        DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC) as length_rank
    FROM film f
    JOIN film_category fc ON f.film_id = fc.film_id
    JOIN category c ON fc.category_id = c.category_id
)
SELECT category_name, title, length, length_rank
FROM RankedFilms
WHERE length_rank <= 3;
```

Data Output | Messages | Notifications

Showing rows: 1 to 64

	category_name character varying (25)	title character varying (255)	length smallint	length_rank bigint
1	Action	Worst Banger	185	1
2	Action	Darn Forrester	185	1
3	Action	Casualties Encino	179	2
4	Action	Quest Mussolini	177	3
5	Animation	Pond Seattle	185	1
6	Animation	Gangs Pride	185	1
7	Animation	Theory Mermaid	184	2
8	Animation	Sons Interview	184	2
9	Animation	Lawless Vision	181	3
10	Children	Wrong Behavior	178	1

Lab 2: Growth Analysis (Time Series)

Result

Total revenue per month (using `DATE_TRUNC`)

Query | Query History

```sql
SELECT
    DATE_TRUNC('month', payment_date) AS payment_month,
    SUM(amount) AS current_revenue
FROM payment
GROUP BY DATE_TRUNC('month', payment_date)
ORDER BY payment_month;
```

Data Output | Messages | Notifications

SQL

	payment_month timestamp without time zone	current_revenue numeric
1	2007-02-01 00:00:00	8351.84
2	2007-03-01 00:00:00	23886.56
3	2007-04-01 00:00:00	28559.46
4	2007-05-01 00:00:00	514.18

MoM Growth % (using `LAG` to retrieve previous month's revenue):

Query | Query History

```sql
WITH MonthlyRevenue AS (
    SELECT
        DATE_TRUNC('month', payment_date) AS payment_month,
        SUM(amount) AS current_revenue
    FROM payment
    GROUP BY DATE_TRUNC('month', payment_date)
)
SELECT
    payment_month,
    current_revenue,
    LAG(current_revenue, 1) OVER (ORDER BY payment_month) AS prev_revenue,
    ROUND(
        (current_revenue - LAG(current_revenue, 1) OVER (ORDER BY payment_month))
        / LAG(current_revenue, 1) OVER (ORDER BY payment_month) * 100
    , 2) AS growth_percentage
FROM MonthlyRevenue
ORDER BY payment_month;
```

Data Output | Messages | Notifications

Showing rows:

	payment_month timestamp without time zone	current_revenue numeric	prev_revenue numeric	growth_percentage numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value (Running Total)

Result

Cumulative sum of transaction amounts per customer (using `SUM()` with `PARTITION BY` and `ORDER BY`):

Query | Query History

```sql
SELECT
    customer_id,
    payment_date,
    amount,
    SUM(amount) OVER (
        PARTITION BY customer_id
        ORDER BY payment_date
    ) AS lifetime_value
FROM payment
ORDER BY customer_id, payment_date;
```

Data Output | Messages | Notifications

Showin

	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	lifetime_value numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91
10	1	2007-03-02 16:30:04.996577	0.99	41.90
11	1	2007-03-17 11:06:20.996577	4.99	46.89
12	1	2007-03-18 02:25:55.996577	0.99	47.88

Lab 4: The ”Top-N” Filter Challenge

Query

Query | Query History

```sql
WITH CustomerSpend AS (
    SELECT
        co.country,
        c.customer_id,
        c.first_name || ' ' || c.last_name AS customer_name,
        SUM(p.amount) AS total_spend
    FROM customer c
    JOIN address a ON c.address_id = a.address_id
    JOIN city ci ON a.city_id = ci.city_id
    JOIN country co ON ci.country_id = co.country_id
    JOIN payment p ON c.customer_id = p.customer_id
    GROUP BY co.country, c.customer_id, c.first_name, c.last_name
),
RankedCustomers AS (

    SELECT
        country,
        customer_name,
        total_spend,
        RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) as spend_rank
    FROM CustomerSpend
)

SELECT
    country,
    customer_name,
    total_spend
FROM RankedCustomers
WHERE spend_rank = 1
ORDER BY country;
```

Result

Top 1 highest-spending customer per country (using CTE with `RANK()` and outer `WHERE rank = 1`):

Data Output | Messages | Notifications

	country character varying (50)	customer_name text	total_spend numeric
1	Afghanistan	Vera Mccoy	67.82
2	Algeria	June Carroll	151.68
3	American Samoa	Anthony Schwab	47.85
4	Angola	Claude Herzog	93.80
5	Anguilla	Bobby Boudreau	99.68
6	Argentina	Jordan Archuleta	129.71
7	Armenia	Stephanie Mitch...	118.75
8	Austria	Nora Herrera	113.74
9	Azerbaijan	Raymond Mcwh...	108.75
10	Bahrain	Seth Hannon	108.76
11	Bangladesh	Michelle Clark	146.68
12	Belarus	Clara Shaw	189.60
13	Bolivia	Joel Francisco	92.78
14	Brazil	Marion Snyder	194.61
15	Brunei	Lois Butler	107.66
16	Bulgaria	Tyrone Asher	112.76
17	Cambodia	Elmer Noe	92.76
18	Cameroon	Albert Crouse	96.78
19	Canada	Curtis Irby	167.62
20	Chad	Javier Elrod	122.72
21	Chile	Janet Phillips	124.74